Technical Report

Limestone Calcined Clay Cement (LC3)

A technical assessment of LC3's raw materials, production processes, hydration mechanisms and potential for low carbon concrete construction

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1. Introduction

Limestone Calcined Clay Cement (LC3) is an innovative blended cement designed to significantly reduce the embodied carbon of concrete by replacing large proportions of clinker, used in traditional Ordinary Portland Cement (OPC), with calcined clay and limestone, both of which are abundant and inexpensive. LC3 sees CO_2 reductions of up to 40% when compared to OPC [1], without compromising strength and durability.

This report examines the raw materials and manufacturing route of LC3, explains its hydration mechanisms and resulting performance, and assesses its potential to support mainstream low carbon concrete construction over the next 20 years.

2. LC3 cement

2.1 Raw materials

LC3 typically contains, 50% clinker, 30% calcined clay, 15% limestone and 5% gypsum.

The clinker provides alite and belite which hydrate to provide early and long-term strength. Kaolinitic clays containing 40–60% kaolinite provides reactive alumina and silica once calcined. Finely ground limestone ($CaCO_3$) acts as a source of carbonate ions needed to form carboaluminates, and a small addition of gypsum ($CaSO_4 \cdot 2H_2O$) regulates early hydration of the clinker phases by controlling C_3A dissolution, therefore aiding the workability of the cement [2].

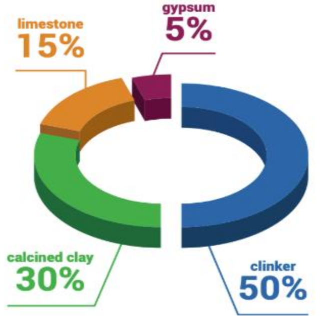

Figure 1:Material breakdown of LC3 [3]

2.2 Manufacturing process

The LC3 manufacturing process is designed to integrate with existing cement plant infrastructure, allowing adoption with minimal modification. The key stages are shown in figure 2 and described below:

2.2.1 Quarrying and preparation

Limestone, Gypsum and Clay are quarried using conventional extraction methods. The materials are crushed and homogenised prior to further processing.

2.2.2 Clay calcining

The kaolinitic clay is heated to approximately 800°C in either a flash calciner or rotary kiln. During heating the Kaolinite ($Al_2Si_2O_5(OH)_4$) undergoes endothermic dehydroxylation to produce metakaolin ($Al_2Si_2O_7$) [1]:

$$Al_2Si_2O_5(OH)_4 \rightarrow Al_2Si_2O_7 + 2H_2O$$

The clay is then rapidly cooled to prevent recrystallisation.

2.2.3 Limestone addition

The limestone in LC3 requires no thermal treatment and is added in its ground form.

2.2.5 Blending

The clinker, calcined clay, limestone, gypsum, and any additional supplementary cementitious materials (SCMs) are proportioned and ground together producing the homogenous cement. Grinding is optimised such that constituents reach desired fineness for efficient reactivity.

2.2.6 Storage and dispatch

The finished LC3 behaves similarly to OPC in terms of handling, conveying and storage. Therefore, it can be transported using existing cement supply chains.

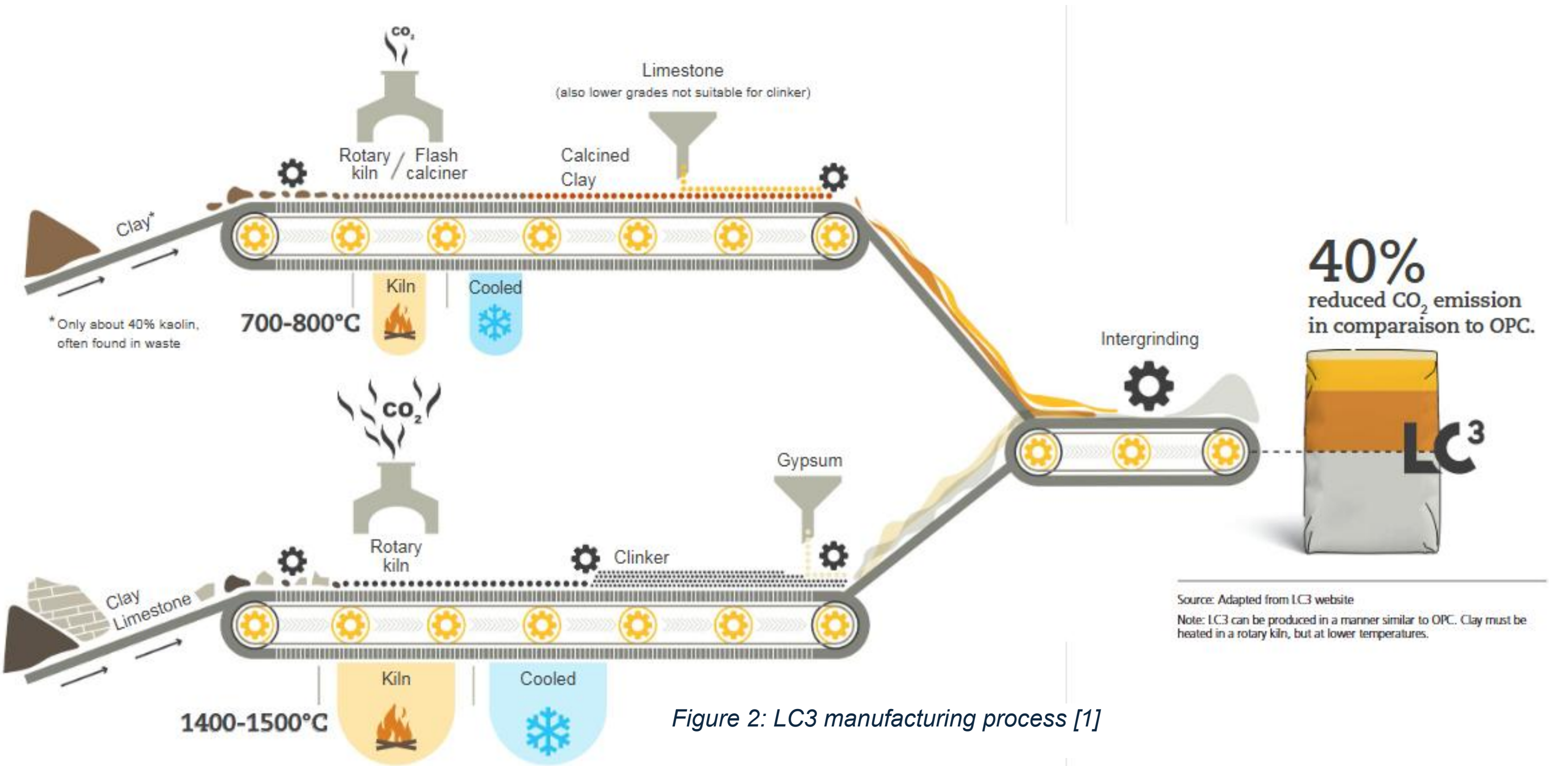

Figure 2: LC3 manufacturing process [1]

2.3 Emissions

LC3 has repeatedly been shown to achieve whole life CO_2 reductions of approximately 40% compared to OPC [1]. With one study showing a 48% emission reduction, from 362.9 kg CO_2/m^3 for OPC to 190.3 kg CO_2/m^3 for LC3, alongside a 43% reduction in total energy consumption (2.142 GJ/m^3 to 1.216 GJ/m^3) [4]. These saving arise due to several factors as shown in figure 3, with key methods listed below:

2.3.1 Reduced clinker content

The primary reduction is due to the clinker content decreasing from 95% in OPC to 50% in LC3, which lowers both process emissions and fuel related emissions.

2.3.2 Lower calcination temperature for clay

Clay is calcined at 800°C this requires less than half the thermal energy required for clinker production at 1400°C [5]. This results in lower fuel use and associated emissions.

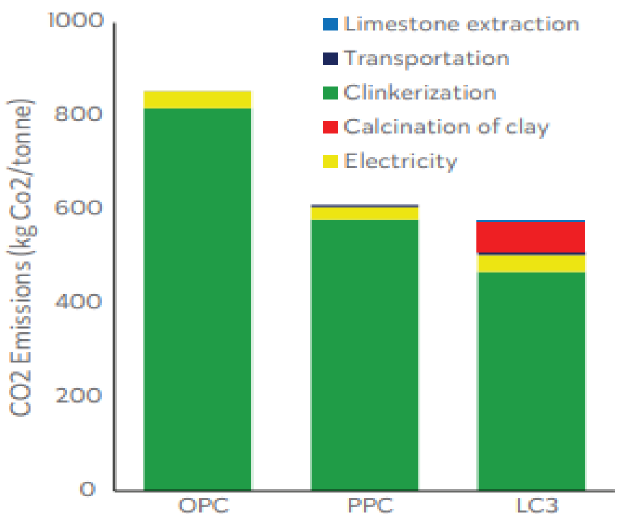

Figure 3: Comparison of C02 emissions of OPC, PPC and LC3 by emission source [6]

2.3.3 Use of low-grade materials

LC3 can use limestone with low carbonate content (≈65%) and natural impurities [5], enabling the use of previously disregarded deposits. It is believed that the amount of cement produced from the same limestone reserve could double [7], reducing demand for new quarrying and the upstream scope 3 emissions linked to extraction, transport, and processing.

2.3.4 Reduced grinding energy

Calcined clay and limestone are significantly softer than clinker. Meaning that LC3 requires significantly less electrical energy during the grinding stage [6]

2.3.5 Avoidance of scarce SCMs

LC3 uses abundant raw materials rather than industrial by-products such as fly ash or ground granulated blast-furance slag (GGBS). This reduces transport distances and associated emissions.

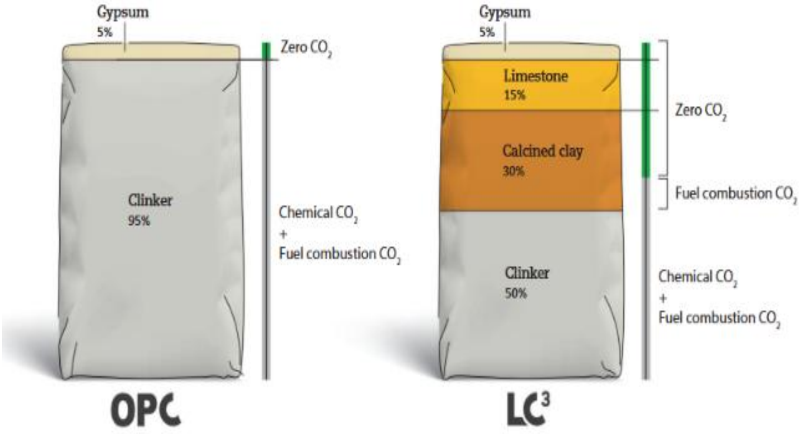

Figure 4: Comparison of OPC and LC3 compositions and associated C02 sources [1]

3. LC3 concrete

3.1 Hydration chemistry

Hydration of LC3 occurs through several interlinked chemical pathways:

3.1.1 Clinker hydration

The Alite (C_3S) and Belite (C_2S) phases hydrate to form calcium silicate hydrate (C-S-H) the primary binding phase of cement, and portlandite (CH) [8].

$$C_3S + 5.3H \rightarrow C_{1.7}SH_4 + 1.3CH$$

$$C_2S + 4.3H \rightarrow C_{1.7}SH_4 + 0.3CH$$

3.1.2 Pozzolanic reaction of calcined clay

The reactive alumina and silica from the metakaolin (AS_2) react with the CH released from clinker hydration to produce calcium-alumino-silicate-hydrate (C-A-S-H) [9] and strätlingite (C_2ASH_8). This refines the pore structure replacing large CH crystals with finer C-A-S-H gel.

$$AS_2 + 3CH + 6H \rightarrow \text{C-A-S-H} + C_2ASH_8$$

3.1.3 Formation of carboaluminate phases

In OPC, alluminates react with gypsum to form ettringite during early hydration. As sulphate depletes, ettringite converts to monosulpahte. In LC3 however the additional limestone promotes the formation of carboaluminate phases (monocarboaluminate ($C_4A_CH_{11}$) and hemi carboaluminate($C_4A_{c_{0.5}}H_{12}$)) [9] These form from the reaction of tricalcium aluminate (C_3A) released from metakaolin, with dissolved carbonate ions from limestone. These carboaluminate phases stabilise ettringite and inhibit its conversion to monosulphate [10]. Maintaining ettringite is beneficial as it occupies a greater solid volume which helps to densify the microstructure, refine pores and therefore improve the durability and performance of LC3.

$$C_3A + C_C + 11H \rightarrow C_4A_CH_{11}$$

$$C_3A + 0.5C_C + 0.5CH + 11.5H \rightarrow C_4A_{c_{0.5}}H_{12}$$

$$AS_2 + 0.5C_C + 3.5CH + 8.5H \rightarrow C_4A_{c_{0.5}}H_{12}$$

3.2 Mechanical performance

laboratory and industrial trails consistently show LC3 achieves strengths comparable to and often better than OPC [11]. Figure 5 shows, although LC3 has slightly lower strength in days 1-3, after 7 days it already outperforms OPC and in days 28-90 shows significantly higher compressive strengths. This superior performance primarily arises from the LC3s denser microstructure. Figure 5 also highlights how LC3 outperforms other SCM blends, such as Slag and Fly-ash based cements, at all curing ages.

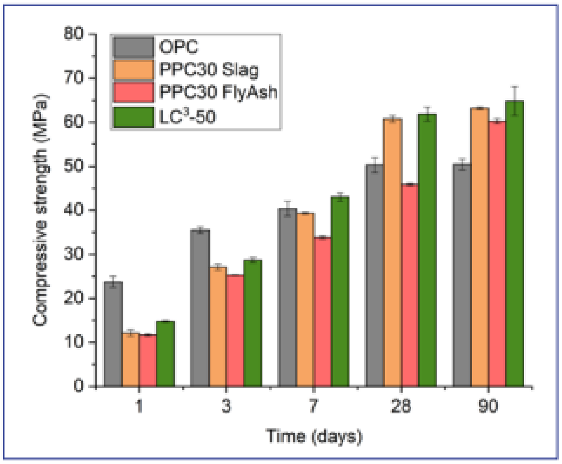

Figure 5: Strength development comparison of LC3, OPC and SCM blends [11]

3.3 Durability performance

LC3 consistently demonstrates durability better than OPC and other SCMs, in both laboratory testing and long-term field exposure.

3.3.1 Surface resistivity

Surface resistivity is a strong indicator of permeability. As shown in figure 6, LC3 exhibits resistivity an order of magnitude higher than OPC and PPC. This indicates a far less permeable pore structure which is consistent with the improved durability performance observed in later sections.

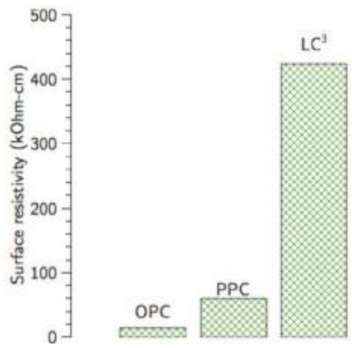

Figure 6: Surface resistivity of LC3, OPC and PPC [12]

3.3.2 Chloride ingress resistance

Chloride ingress is a critical factor in the durability of reinforced concrete [13]. As shown in figure 7, although some LC3 mixes may exhibit higher chloride contents at surface level, their profiles decay much more rapidly. In all cases chloride content fell below OPC within 10mm and with reinforcement normally at ≥25 mm cover, LC3 offers better corrosion protection.

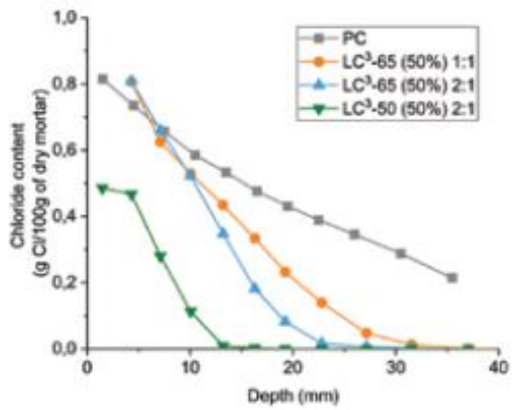

Figure 7: Total chloride content versus depth for OPC and LC3 blends [13]

Long term marine exposure tests, shown in figure 8, align with laboratory results. Showing LC3 blocks remaining corrosion free, whereas OPC blocks exhibited extensive deterioration [14].


Figure 8: Field exposure of LC3 and OPC concrete blocks [14]

3.3.3 Alkali Silica Reaction resistance

ASR occurs when alkalis in the pore solution react with silica in aggregates, forming an expansive gel that leads to cracking. As shown in figure 9, LC3 exhibits exceptionally low expansion (0.04% after 1 year) compared to 0.5% for OPC. The higher levels of alumina inhibit the dissolution of silica and supresses ASR gel formation [15].

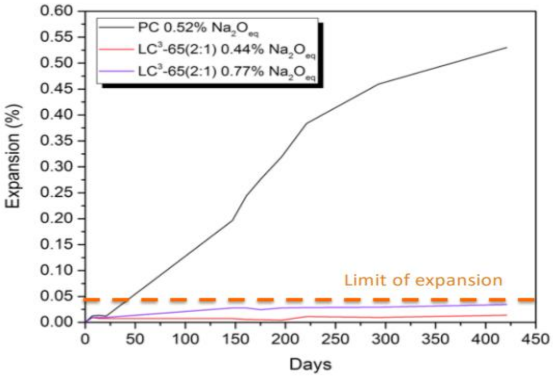

Figure 9: Expansion behaviours of LC3 and OPC showing relative amount of Alkali Silica Reaction occurring [9]

4. Adoption of LC3 cement

4.1 Drivers for adoption

The key driver for LC3s use is its reduction in CO_2 emissions as discussed in section 2, it provides a practical route for producers seeking compliance as carbon taxation and emissions regulations become more stringent and widely implemented.

LC3 offers several advantages that support rapid industrial uptake. A major driver is its compatibility with existing cement plant infrastructure: around 75% of existing cement plants can produce LC3 with only minor adjustments [1], and no additional specialist training is required.

As discussed in the previous section LC3 shows favourable mechanical and durability performances, which allows for its use in a wide array of projects.

As figure 10 shows, a major advantage LC3 has over other SCMs is the availability of raw materials. Kaolinitic clay and low-grade limestone deposits occur globally, while by-products such as GGBS and fly ash's availability is decreasing due to decarbonisation of the power and steel sectors. This abundance ensures long term supply security.

LC3 offers strong economic advantage, with production cost reductions of 25% compared to OPC [1], and even greater savings in clinker importing regions. These savings make LC3 attractive to producers.

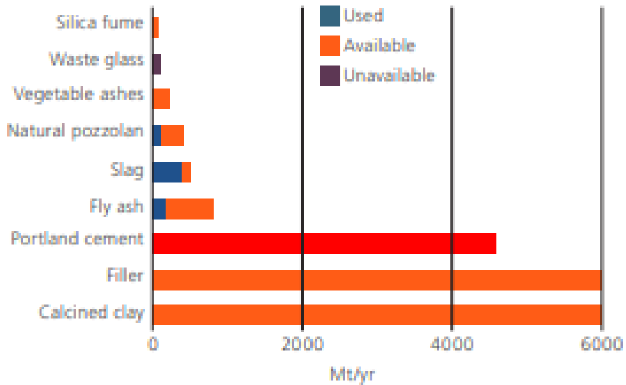

Figure 10: Global use and estimated availability of SCMs and fillers [16]

4.2 Barriers to adoption

Despite its advantages, several challenges limit uptake. Although abundant on a global scale, some regions have limited deposits of kaolinic rich clay, and may need to import, reducing economic and environmental benefits.

Technical challenges arise due to calcined clay's high surface area increasing water demand [17]. Without the use of modern superplasticisers, workability is reduced [18]. Another limitation is colour control, clay often creates a reddish cement which may be undesirable for architectural concrete. While pigments and other recent technologies can mitigate this, they add additional cost and complexity.

A significant barrier is the absence of design codes and standards for LC3. Most standards are based on OPC or established SCMs, until dedicated LC3 based standards and performance specifications are adopted its use is likely to remain restricted.

4.3 Mainstream adoption

LC3 has progressed rapidly from laboratory research to large scale industrial trials. LC3 has gained significant attention and is recognised as one of the most promising technologies to reduce emissions in the cement sector. This growing industrial interest has more than 40 cement companies in 25 different countries [3] currently involved in LC3 development or production.

Its adoption potential is strengthened by resource distribution, many regions expected to see the fastest growing cement demand particularly Africa, South Asia, and South America, possess large accessible kaolinitic clay deposits. This allows LC3 to be produced close to point of use, reducing transportation costs and emissions, while improving supply resilience.

Economic analysis, as shown in table 1, shows strong financial viability. When LC3 is produced in an existing cement plant (scenario 1) with nearby clay sources, estimated internal rates of return (IRR) exceed 60%. Even when clay needs transporting up to 200km, IRRs remain in the 20-25% range. In new 'greenfield'' plants (Scenario 3) profitability scores lowly due to the large construction cost of a new plant.

LC3 has already been used in residential building, commercial structures, industrial projects, tunnels, and bridges. It has also been applied in masonry blocks, pavers, and precast elements. These projects demonstrate the versatility of LC3 and provide long term performance data.

Training programmes, conferences, and workshops are being run to familiarise engineers, architects, and policymakers with LC3. Ongoing research is generating the data required for adoption into codes, including mapping suitable clay sources, developing testing methods, and validating field and laboratory results. This will facilitate widespread structural use as designers and insurers usually require compliance with recognised codes before new cement types can be specified.

Figure 11: Real world examples of LC3s application [3]

Table 1: Economic feasibility scenarios for LC3 production [13]

Scenario	Calciner type	Capital expenditure (US$m)	Clay availability – distance (km)	LC3 production cost (US$/t)	Attractiveness – IRR (%)
Scenario 1 – 1Mta LC^3 in cement plant (0.3Mt calcined clay)	Flash calciner	10.3	10	23.4	63
			200	27.3	22
	Rotary kiln	6.6	10	24.2	87
			200	28.1	24
Scenario 2 – 0.413Mt LC^3 in grinding unit (0.124Mt calcined clay)	Flash calciner	8.15	10	32.1	75
			200	36.0	55
	Rotary kiln	6.1	10	32.6	98
			200	36.5	71
Scenario 3 – 0.413Mt LC^3 in greenfield project (0.124Mt calcined clay)	Flash calciner	27.0	10	32.1	17
			200	36.0	9
	Rotary kiln	26.0	10	32.6	17
			200	36.5	9

5. Conclusion

LC3 cement offers a practical, scalable, and economically feasible route to reducing the carbon footprint of global cement production. By replacing up to 50% of clinker, LC3 achieves significant CO2 reductions while its hydration mechanisms, produce a refined microstructure that provide strength and durability comparable or better than OPC.

A key advantage is its compatibility with existing cement manufacturing infrastructure and its potential for lowering production costs. Together with growing industry interest, LC3 has a clear pathway to mainstream adoption.

Given the scale of global cement use, any viable low carbon alternatives must be producible at industrial volumes using widely available materials. Unlike many other emerging cements that face limitations in resource availability or specialised processing needs, LC3 satisfies this requirement.

Therefore, I believe LC3 is highly likely to become a mainstream construction material within the next 20 years and will play a significant role in enabling low carbon concrete construction.

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